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Antoine Bonnet

Building computer-use agents, from evals to products.

Experience

Member of Technical Staff, Agents 04/2025 – present
H Company, computer-use agent lab *Paris, France*

- **Agent harness.** Created and lead H's agent harness (60k lines of Python, 59% of commits across 56 contributors). It runs data collection and evals for Holo3, 3.1 and 4, the Agents API, every product and customer deployments. Top contributor to H's monorepo: 1.4k merged PRs among 94 engineers
- **Holo4 (co-author): harness engineering.** Ran a loop where agents diagnose failures and propose harness fixes, each reviewed by hand (60 runs). Took Qwen3.8-27B from 24.8% to 59.8% on OSWorld 2.0 without finetuning, beating Alibaba's reported 48.0%, via memory compaction and a code-only desktop CLI. Holo4 27B: 61.7% at 1/7 Opus 5.5's cost
- **Evals and data.** Brought 20+ benchmarks (OSWorld 2.0, CUA-Gym, HLE, GDPval, BrowseComp) into nightly regression runs, with LLM-judged failure analysis, RL step rewards and cheating audits. Built the converter that turns any agent trace into SFT data. Holo3.5 data collection: prompts 3× shorter, +7 to +14 pts
- **Surfer 2 and Holo2.** Core contributor to Surfer 2 (orchestrator, judges), state of the art on OSWorld (60.1%), WebVoyager (97.1%), WebArena (69.6%), AndroidWorld (87.1%). Built Holo2's agentic localization: 78.5% on ScreenSpot-Pro, a record
- **Agents API (lead engineer).** Shipped H's public computer-use API 10 weeks after the first PR: 72 REST operations (sessions, streaming, webhooks, custom tools), Python and TypeScript SDKs, CLI, MCP server and docs
- **Products.** Created HoloTab, a browser agent in Chrome (6k users, rated 4.9/5). Created HoloBricks and HoloBlocks: describe a model and an agent builds it in LEGO or Minecraft, up to 27k bricks; shipped in 2 weeks with 11 people. Started the open-source HoloDesktop CLI and trajectories.hcompany.ai

Research Scientist Intern 09/2024 – 03/2025
Meta (GenAI) *Paris, France*

- Built a hierarchical multi-agent system where a planner delegates to sub-agents and revises its plan; found that reasoning models beat standard agents with half the tool calls
- Fine-tuned Llama 3.1 70B on its own successful agent runs (rejection sampling): +6.5% over the post-trained baseline

Data Lead, Meditron 06/2023 – 09/2024
MLO & NLP Labs, EPFL *Lausanne, Switzerland*

- Led the data pipeline for *MEDITRON-70B*, a medical Llama 2: built the pretraining corpus and released the *Clinical Guidelines* dataset. 2.2k GitHub stars; featured on the Meta AI blog
- Built *MOOVE*, a clinician feedback platform for medical LLMs: 1,237 annotators, 22 organizations, 18k contributions

Publications and Open Source

- *Holo4*, H Company, 2026 (co-author); also Holotron-12B and Holo2-235B-A22B. Wrote the Holo3.1 launch post
- *Surfer 2: Cross-Platform Computer Use Agents*, 2025 (53 authors, alphabetical)
- *Surfer-H Meets Holo1: Cost-Efficient Web Agent Powered by Open Weights*, 2025 (44 authors, alphabetical)
- *MEDITRON-70B: Scaling Medical Pretraining for LLMs*, Chen, …, **Bonnet*** et al., 2023 (*equal contrib.; 850+ citations)
- Creator of hai-agents-python, hai-agents-ts, holo-desktop-cli; merged X11 fixes to trycua/cua (30k stars)

Education

MSc Computer Science, EPFL 09/2022 – 03/2025
Thesis: Multi-Agent Orchestration via Hierarchical Communication, at Meta GenAI *GPA 5.47/6.00*

BSc Joint Honours Math and Computer Science, McGill 09/2019 – 05/2022
First-Class Honours, GPA 3.92/4.00, NSERC Undergraduate Research Award *Montréal, Canada*

Skills

Agents and ML: tool calling, multi-agent systems, LLM judges, RL environments, SFT data, PyTorch, vLLM, Hugging Face
Engineering: Python, TypeScript, Kubernetes, AWS, Docker, REST/OpenAPI, MCP, Chrome extensions